第五节　第一型的曲面积分

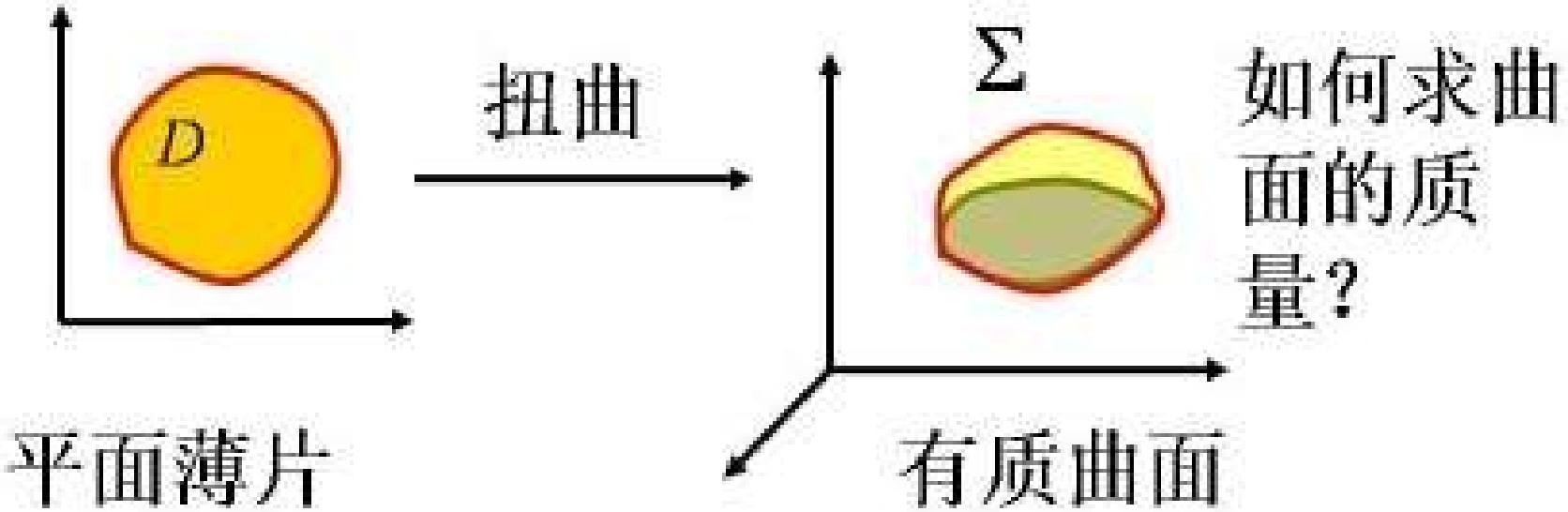
一、第一型曲面积分的概念性质

二、第一型曲面积分的计算

回忆：二重积分可以平面薄片的质量

面密度：$\rho = \rho(x, y)$

$$M = \lim_{\lambda \to 0} \sum_{i=1}^{n} \rho(\xi_i, \eta_i) \Delta\sigma_i = \iint_D \rho(x, y) d\sigma$$



定义　设$f(x, y, z)$是定义在曲面$\sum$上的有界函数,

将$\sum$任意分成n小块 ΔS_i(同时表示其面积)

在每小块上任取一点(ξ_i, η_i, ζ_i)

作和式 $\sum_{i=1}^{n} f(\xi_i, \eta_i, \zeta_i)\Delta S_i$

如果当各小块曲面直径的最大值 $\lambda \to 0$时，这和式的极限存在, 则称此极限值为函数$f(x, y, z)$在曲面$\sum$上对面积的曲面积分或第一类曲面积分, 记为

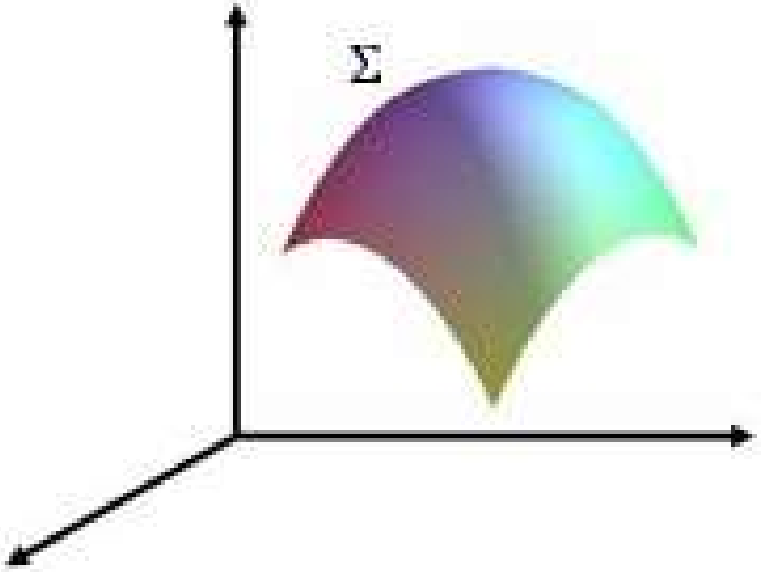
$$\iint_{\Sigma} f(x, y, z)dS = \lim_{\lambda \to 0} \sum_{i=1}^{n} f(\xi_i, \eta_i, \zeta_i)\Delta S_i$$

积分曲面

$$\iint_{\Sigma} f(x,y,z)\,dS = \lim_{\lambda\to 0}\sum_{i=1}^{n} f(\xi_i,\eta_i,\zeta_i)\Delta S_i$$

对面积的曲面积分

第一类曲面积分



注: (1) 若Σ是封闭曲面，则上述积分记为$\oiint\limits_{\Sigma} f(x,y,z)dS$

(2) 如果$f(x, y, z)$在曲面$\sum$上连续，则曲面积分存在

(3) 曲面积分具有与其它积分类似的性质,如

$$\iint\limits_{\Sigma=\Sigma_1+\Sigma_2} f(x,y,z)dS = \iint\limits_{\Sigma_1} f(x,y,z)dS + \iint\limits_{\Sigma_2} f(x,y,z)dS$$

(4)非均匀曲面的质量$M = \iint\limits_{\Sigma} \mu(x,y,z)dS$

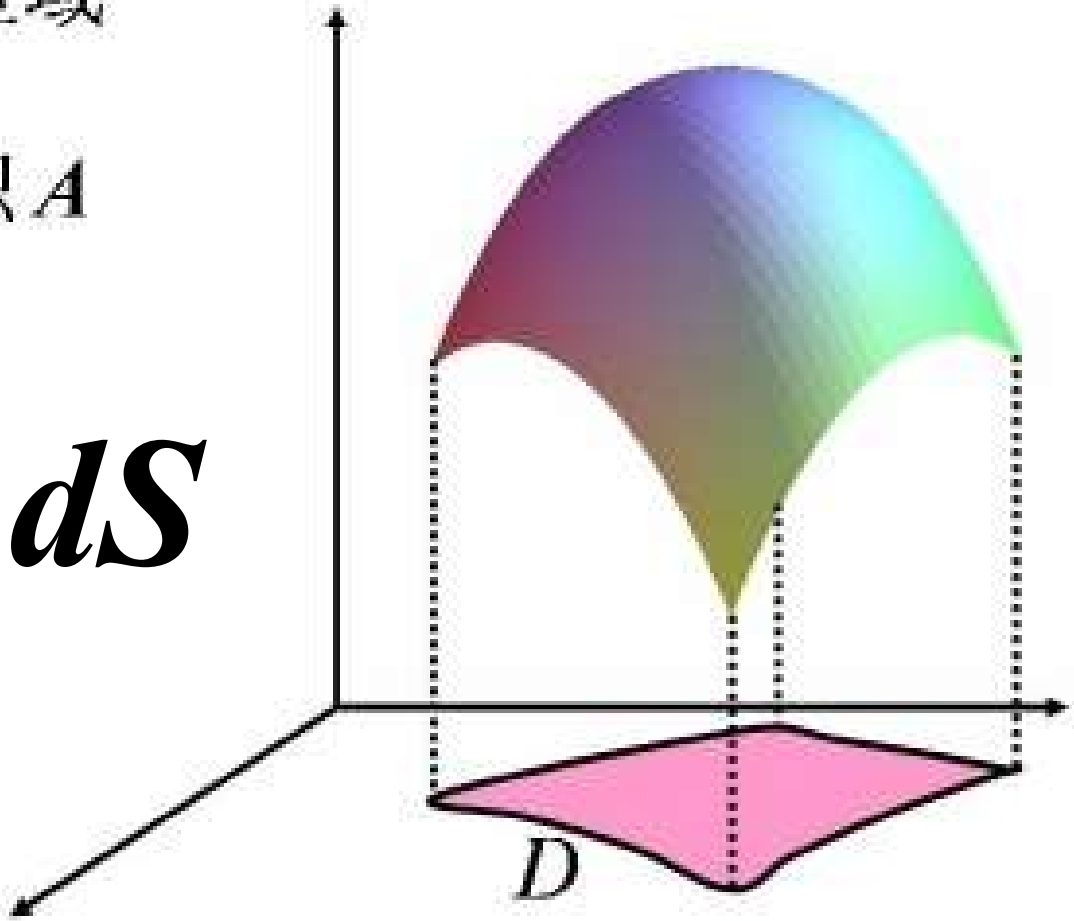
(5) $f(x,y,z)=1, \quad S = \iint\limits_{\Sigma} dS$

曲面方程：$z = f(x, y) \quad (x, y) \in D$

D：有界闭区域

求曲面的面积 A

$$\mathbf{A} = \iint_{\Sigma} dS$$



(6) 对称性:

对面积的曲面积分 $\iint\limits_{\Sigma} f(x,y,z)\,\mathrm{d}S$,

对称性的利用类似于三重积分.

如：若 $f(x,y,z)$ 在 Σ 上连续，Σ 关于 yoz 面对称，则

$$\iint\limits_{\Sigma} f(x,y,z)\,\mathrm{d}S = \begin{cases} 0, & f(-x,y,z) = -f(x,y,z) \\ 2\iint\limits_{\Sigma_1} f(x,y,z)\,\mathrm{d}S, & f(-x,y,z) = f(x,y,z) \end{cases}$$

Σ_1 : Σ 在 $x \geq 0$ 的部分.

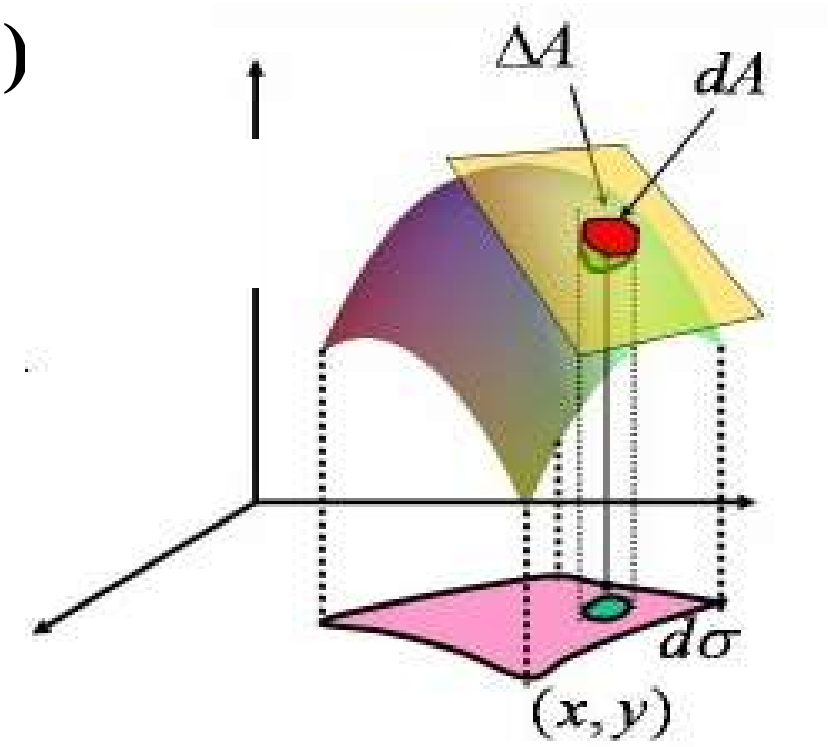
1 设曲面的方程为：$z = f(x, y)$

在 xoy 面上的投影区域为 D,

如图，设小区域 $d\sigma \in D$,

点 $(x, y) \in d\sigma$,

Π 为 Σ 上过 $M(x, y, f(x, y))$ 的切平面 .



以 $d\sigma$ 边界为准线，母线平行 于 z 轴的小柱面，截曲面 Σ 为 ΔA；截切平面 Π 为 dA，则有 $dA \approx \Delta A$.

$$d\sigma = dA \cdot \cos\gamma,$$

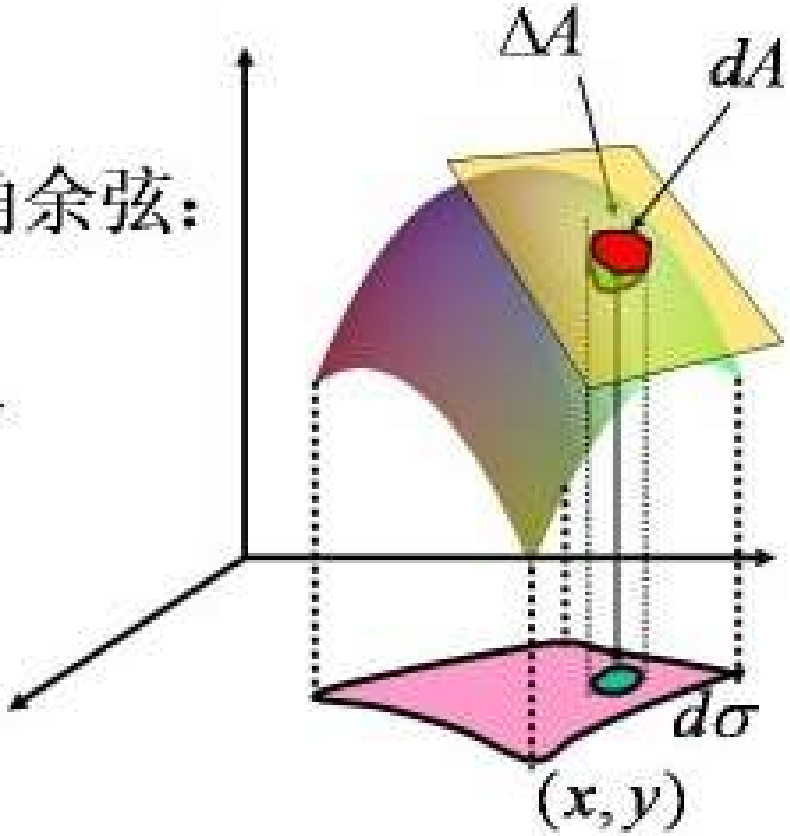
切平面的法向量

$$\mathbf{n}=\{-f_x,-f_y,1\}$$

切平面与 $z=0$ 的夹角余弦：

$$\cos\gamma=\frac{1}{\sqrt{1+f_x^{\,2}+f_y^{\,2}}}$$

（ **n**与 **k** 的夹角余弦 ）

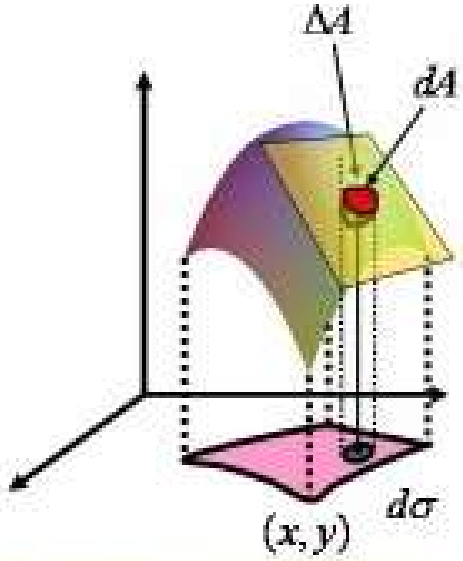


$\because d\sigma$ 为 dA 在 xoy 面上的投影， $\therefore \quad d\sigma = dA \cdot \cos\gamma,$

$$\cos\gamma = \frac{1}{\sqrt{1+f_x^{\,2}+f_y^{\,2}}}$$

$$\therefore dA = \frac{d\sigma}{\cos\gamma}$$

$$\boxed{dA = \sqrt{1+f_x^{\,2}+f_y^{\,2}}\,d\sigma}$$

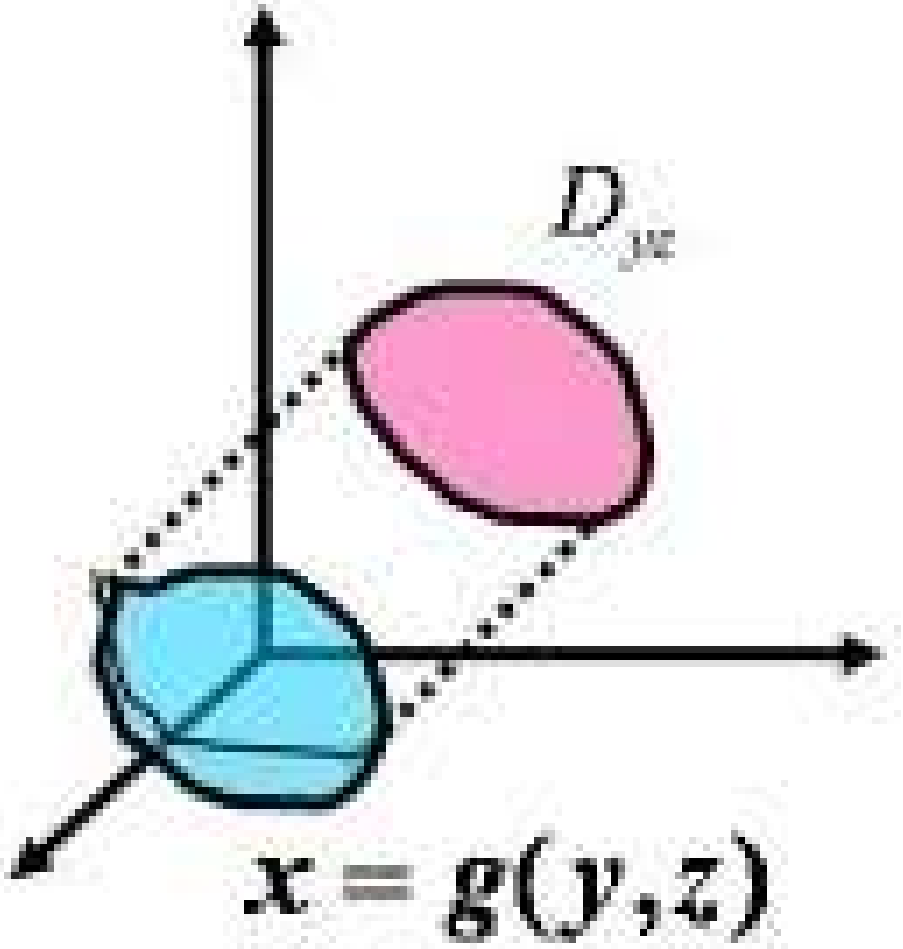


面积元素

$$\therefore dS = \sqrt{1+f_x^2+f_y^2}\,d\sigma,$$

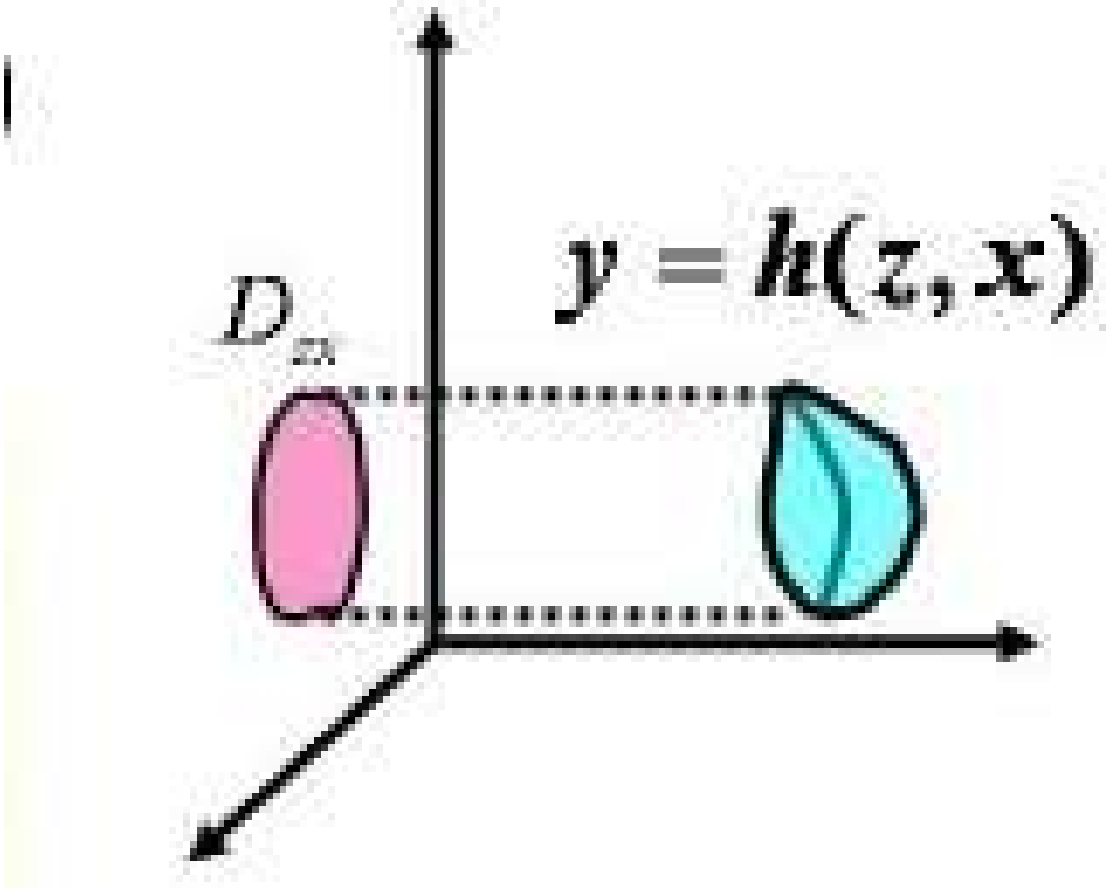
2．设曲面的方程为： $x = g(y,z)$

曲面面积元素为： $$dS = \sqrt{1 + \left(\frac{\partial x}{\partial y}\right)^2 + \left(\frac{\partial x}{\partial z}\right)^2}\, d\sigma$$



3．设曲面的方程为： $y = h(z, x)$

曲面面积元素为： $dS = \sqrt{1 + \left(\frac{\partial y}{\partial z}\right)^2 + \left(\frac{\partial y}{\partial x}\right)^2}\, d\sigma.$

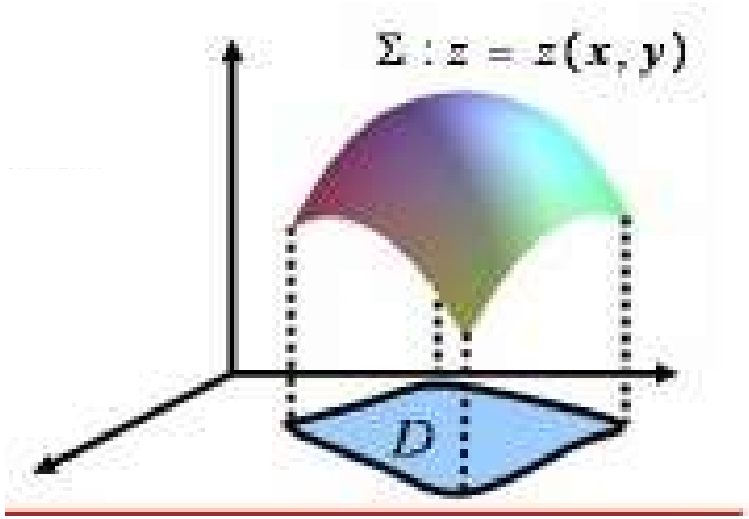


二、对面积的曲面积分的计算法

化成二重积分

1.设曲面∑的方程为$z=z(x,y)$

∑在xoy面上的投影区域为 D_{xy}



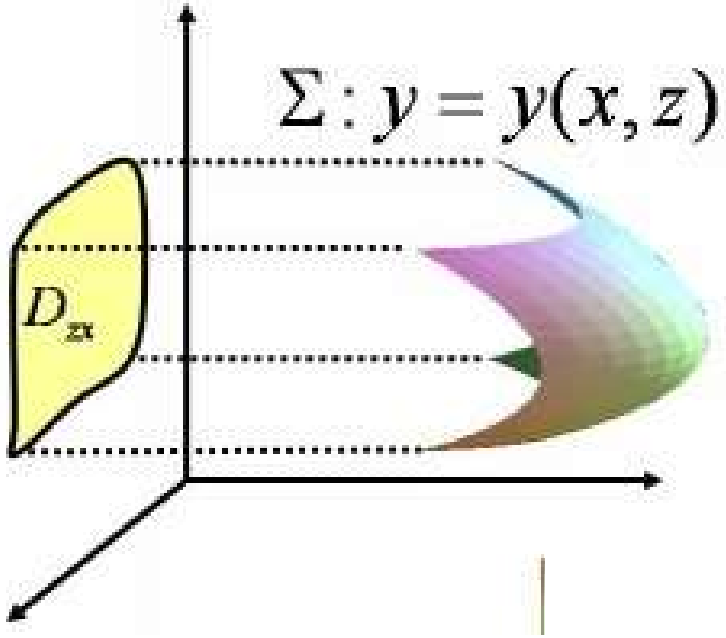
$$dS=\sqrt{1+z_x^2+z_y^2}\,dxdy$$

$$\iint_{\Sigma} f(x,y,z)dS$$

$$=\iint_{D_{xy}} f[x,y,z(x,y)]\sqrt{1+z_x'^{\,2}+z_y'^{\,2}}\,dxdy;$$

2.设曲面∑的方程为 $y=y(z,x)$

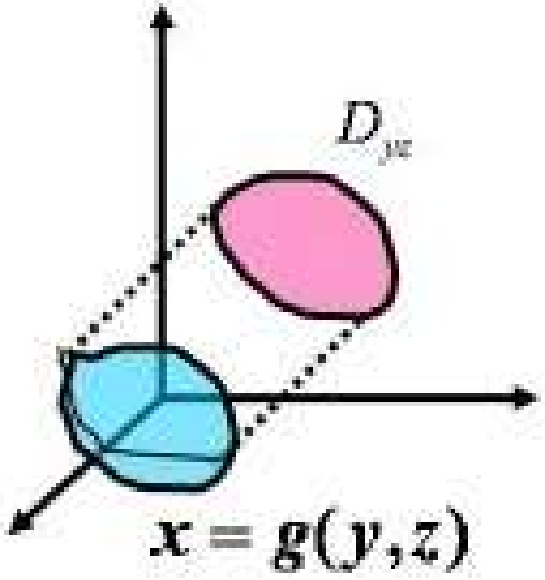
∑在zox面上的投影区域为 D_{zx}



$$\iint_{\Sigma} f(x,y,z)dS = \iint_{D_{zx}} f[x,y(z,x),z]\sqrt{1+y_x^2+y_z^2}\,dzdx$$

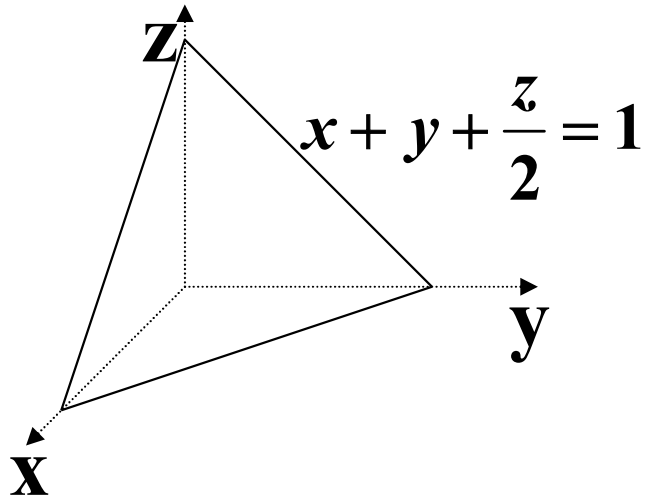
3. 若曲面∑的方程为 $x=x(y,z)$

∑在yoz面上的投影区域为 D_{yz}



则 $$\iint_{\Sigma} f(x,y,z)dS = \iint_{D_{yz}} f[x(y,z),y,z]\sqrt{1+x_y^2+x_z^2}\,dydz$$

例1 计算 $\iint\limits_{\Sigma} yzdS$



解: $\Sigma \ : z = 2(1-x-y),$

$D_{xy} : 0 \le x \le 1, 0 \le y \le 1-x$

$z_x = -2, \quad z_y = -2,$

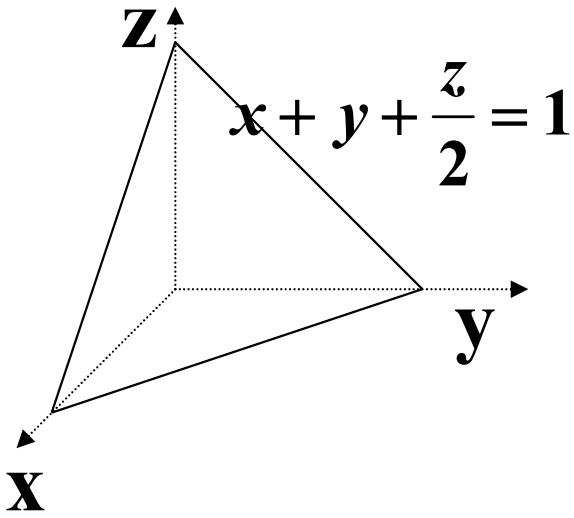
$dS = \sqrt{1+4+4} = 3dxdy$

$$\therefore \iint\limits_{\Sigma} yzdS = \iint\limits_{D_{xy}} y2(1-x-y)3dxdy$$

$$= 6\int_0^1 dx \int_0^{1-x} y(1-x-y)dxdy$$

$$\Sigma \ : y = 1 - x - \frac{z}{2},$$

$$D_{xz} : 0 \le x \le 1, 0 \le z \le 2(1-x)$$



$$\iint_{\Sigma} yz dS = \iint_{D_{zx}} (1 - x - \frac{z}{2}) z \sqrt{1 + y_x^2 + y_z^2} dz dx$$

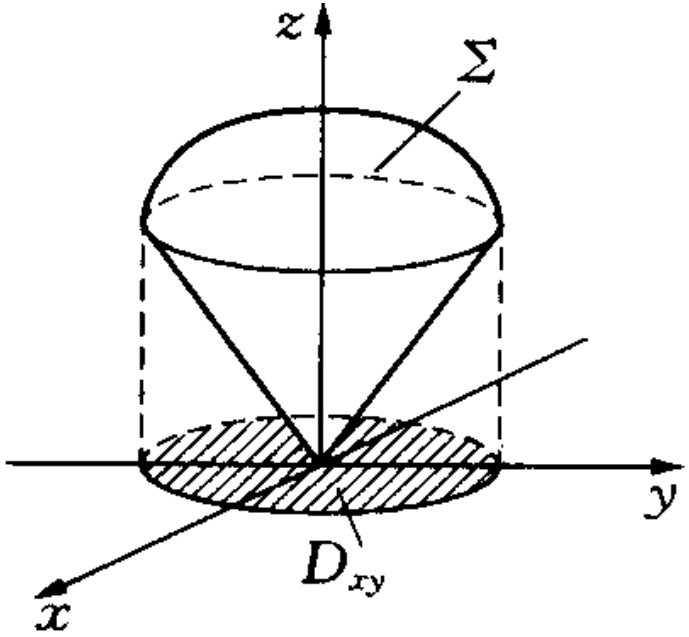
$$\Sigma \ : x = 1 - y - \frac{z}{2},$$

$$D_{yz} : 0 \le y \le 1, 0 \le z \le 2(1-y)$$

$$\iint_{\Sigma} yz dS = \iint_{D_{zx}} yz \sqrt{1 + x_y^2 + x_z^2} dz dy$$

例2.计算 $\iint_{\Sigma} z^3 dS$ 其中 Σ 是上半球面 $z=\sqrt{a^2-x^2-y^2}\,(a>0)$

在圆锥面 $z=\sqrt{x^2+y^2}$ 以上的部分.



解: 两曲面交线在xOy面上的投影为:

$$\begin{cases} z=\sqrt{a^2-x^2-y^2} \\ z=\sqrt{x^2+y^2} \end{cases}, \quad x^2+y^2=\frac{a^2}{2},$$

$$D_{xy}: \ x^2+y^2 \le \frac{a^2}{2},$$

$$dS=\sqrt{1+(z'_x)^2+(z'_y)^2}\,dxdy=\sqrt{1+(\frac{-x}{z})^2+(\frac{-y}{z})^2}\,dxdy$$

$$=\frac{a}{\sqrt{a^2-x^2-y^2}}dxdy$$

故 $\iint_{\Sigma} z^3 dS=\iint_{D_{xy}}(a^2-x^2-y^2)^{\frac{3}{2}}\frac{a}{\sqrt{a^2-x^2-y^2}}dxdy$

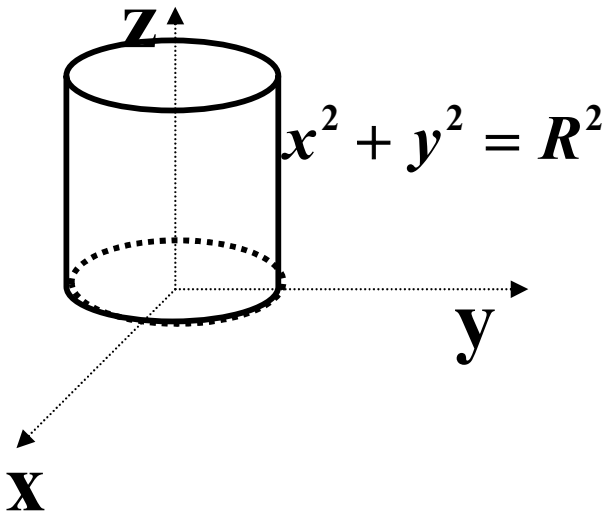
$$= \quad a\iint_{D_{xy}}(a^2 - x^2 - y^2)dxdy$$

$$= \quad a\int_0^{2\pi} d\vartheta \int_0^{\frac{a}{\sqrt{2}}}(a^2 - \rho^2)\rho d\rho$$

$$= 2\pi a[-\frac{1}{4}(a^2 - \rho^2)^2]_0^{\frac{a}{\sqrt{2}}} = 2\pi a \times \frac{3a^4}{16}$$

$$= \frac{3}{8}\pi a^5.$$

例3.计算曲面积分$\iint\limits_{\Sigma}\frac{1}{x^2+y^2+z^2}\mathrm{d}S$



解： $\Sigma_1: x=\sqrt{R^2-y^2}$

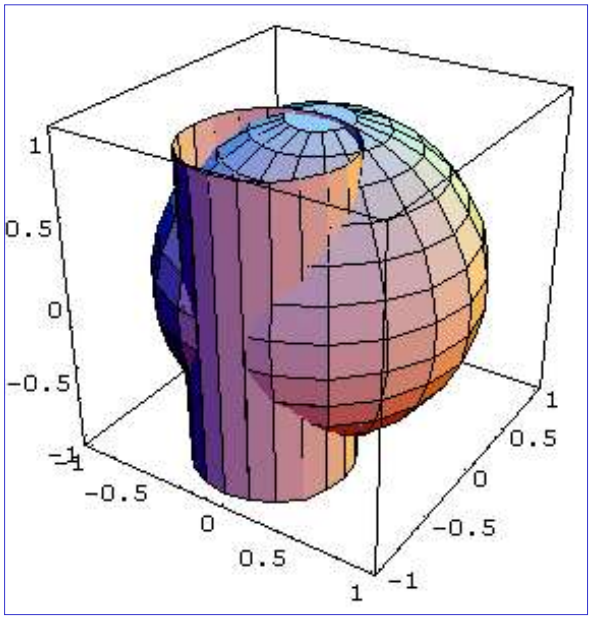
$$D_{zy}: -R\le y\le R, 0\le z\le H$$

$$\Sigma_2: x=-\sqrt{R^2-y^2}$$

$$D_{zy}: -R\le y\le R, 0\le z\le H$$

$$\iint\limits_{\Sigma}=\iint\limits_{\Sigma_1}+\iint\limits_{\Sigma_2}=2\iint\limits_{D_{yz}}f[x(y,z),y,z]\sqrt{1+x_y^2+x_z^2}\,dydz$$

例 4 求球面 $x^2+y^2+z^2=a^2$，含在圆柱体 $x^2+y^2=ax$ 内部的那部分面积.



解 由对称性知 $A=4A_1$,

D_1： $x^2+y^2\le ax \quad (x,y\ge 0)$

曲面方程 $z=\sqrt{a^2-x^2-y^2}$,

于是 $\sqrt{1+\left(\frac{\partial z}{\partial x}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2}$

$$=\frac{a}{\sqrt{a^2-x^2-y^2}},$$

$$A = 4\iint_{\Sigma_1} dS$$

$$A = 4\iint_{D_1} \sqrt{1 + z_x^2 + z_y^2}\, dxdy$$

$$= 4\iint_{D_1} \frac{a}{\sqrt{a^2 - x^2 - y^2}}\, dxdy$$

$$= 4a\int_0^{\frac{\pi}{2}} d\theta \int_0^{a\cos\theta} \frac{1}{\sqrt{a^2 - \rho^2}}\, \rho d\rho$$

$$= 2\pi a^2 - 4a^2.$$

例 $\oiint_{\Sigma}(x+y+z)dS \quad \Sigma: x^2+y^2+z^2=a^2$

例 $\oiint_{\Sigma}x^2dS \quad \Sigma: x^2+y^2+z^2=a^2$

1 变密度细棒的质量 $\int_a^b f(x)dx$

2 变密度平面薄片的质量 $\iint_D f(x,y)d\sigma$

3 变密度空间立体的质量 $\iiint_\Omega f(x,y,z)dv$

4 变密度平面曲线的质量 $\int_L f(x,y)ds$

5 变密度空间曲线的质量 $\int_L f(x,y,z)ds$

6 变密度空间曲面的质量 $\iint_\Sigma f(x,y,z)dS$